Cransfield Materials Testing Center
Quality Assurance

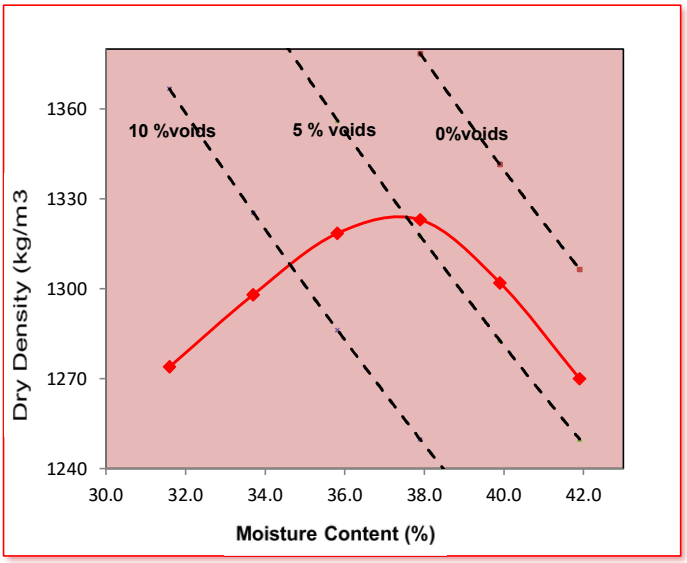
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DENSITY/MOISTURE CONTENT RELATIONSHIP (BS 1377 PART- 4, 3.3 : 1990)

Client name:	**STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT**					
Project/Site Name:	**PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE**					
Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	13-Jun-26	
Sample ID:	BH 1	**Sample depth (M):**	0.0-25.0	**Location:**		

Moisture addition	PMC	50CC	200CC	250CC	300CC	350CC
Wt of mould + wet material (g)	4564	4617	4667	4697	4695	4677
Wt of mould + Base (g)	3050	3050	3050	3050	3050	3050
Wt wet material (g)	1514	1567	1617	1647	1645	1627
Volume of mould (m^3)	0.903	0.903	0.903	0.903	0.903	0.903
Bulk density (kg/m^3)	**1677**	**1735**	**1791**	**1824**	**1821**	**1802**

Container No	DT	6Y	FD	JI	JI	OL
Wt of container + wet material (g)	204.4	169.0	201.0	201.2	204.1	256.3
Wt dry material (g)	155.3	126.4	148.0	145.9	145.9	180.6
Wt of moisture (g)	49.07	42.60	53.00	55.30	58.21	75.67
Wt of container (g)	-	-	-	-	-	-
Moisture content (%)	**31.6**	**33.7**	**35.8**	**37.9**	**39.9**	**41.9**
Dry density (kg/m^3)	**1274**	**1298**	**1319**	**1323**	**1302**	**1270**



Maximum Dry Density (kg/m^3):	**1325**	**Optimum Moisture Content (%):**	**37.6**
Bulk Density (kg/m^3):	**1823**		

Tested by:	CRANSFIELD	**Date reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
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Sign: